

Sophia Li

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Profile

Mechanical Engineering and Aeronautics and Astronautics student at Stanford University with hands-on experience in CAD, structural analysis, composite manufacturing, finite element analysis, and full-cycle hardware design for aerospace systems. Experienced in translating engineering concepts from analysis through manufacturing and testing. Passionate about developing reliable, high-performance mechanical systems for aerospace and defense applications.

Education

Mechanical Engineering, Stanford University

Palo Alto, CA | 09/2025 – 06/2029

- GPA: 4.048
- Relevant Coursework:
 - CS223A/ME320 – Introduction to Robotics
 - CS109 – Probability for Computer Scientists
 - AA100 – Introduction to Aeronautics and Astronautics
 - CS106A, CS106B – Programming (Python and C++)
 - CME102 – Ordinary Differential Equations for Engineers

Skills

- **CAD & FEA:** Autodesk Fusion 360, Abaqus (FEA), HyperX (Collier Aerospace)
- **Analysis:** Classical Laminate Theory, Composite Structures, Structural Analysis, Mechanics of Materials, Failure Criteria
- **Aerodynamics:** XFLR5, Airfoil Analysis, Lift Distribution Modeling, Wing Sizing
- **Hardware:** Rapid Prototyping, Flight Hardware Development, Design for Manufacturing (DFM)
- **Manufacturing:** Composite Manufacturing, 3D Printing, Soldering, Woodworking, Circuit Wiring, Fabrication Tooling
- **Programming:** Python (CS106A, CS109), C++ (CS106B), MATLAB (CME100, CME102)

Engineering Projects

Composite Wing Design, Analysis, Manufacturing, & Testing

06/2026 – 07/2026

- Designed and manufactured a carbon fiber composite foam-core NACA 2412 wing structure, completing the full structural design cycle from aerodynamic loading to physical prototype.
- Built 2D/3D Abaqus composite shell models, validating structural response within 2% of analytical predictions.
- Developed MATLAB-based structural analysis tools to evaluate laminate stiffness, neutral axis location, and bending response, informing composite layup and spar design decisions.
- Performed structural failure analysis using Tsai-Wu criteria to identify laminate failure modes and improve composite design margins.
- Fabricated flight-representative composite wing hardware using wet layup and foam-core manufacturing techniques, enabling experimental validation of structural models.
- Conducting structural verification testing to validate analytical and FEA predictions.

Custom Hot Wire Cutter Design & Fabrication

07/2026

- Designed and fabricated a nichrome wire hot wire cutting system for precision aerospace foam-core manufacturing.

- Engineered a tensioning mechanism to maintain straight cutting geometry across long-span foam sections.
- Integrated electrical power delivery and thermal control considerations for consistent foam cutting performance.
- Applied knowledge of electrical systems, heat transfer, mechanical design, and fabrication processes to develop a custom manufacturing tool.
- Enabled rapid prototyping of aerodynamic foam cores for composite aircraft structures.

Experience

Mechanical Design Lead, Stanford Design/Build/Fly

Stanford University | 09/2025 – Present

- Lead mechanical design and prototyping for AIAA competition aircraft, managing 10 engineers across CAD, manufacturing, and testing phases.
- Translated aircraft performance requirements into manufacturable mechanical subsystem designs using CAD, structural reasoning, and rapid prototyping workflows.
- Designed, fabricated, and tested a two-stage servo-actuated mechanical release mechanism, integrating CAD, electrical components, and flight hardware requirements.
- Managed full aircraft CAD integration in Fusion 360, coordinating fuselage, wing, and control surfaces
- Created manufacturing documentation and supported design reviews to accelerate prototype development and fabrication cycles.

Research Assistant, Stanford Freeman Spogli Institute (FSI)

Stanford University | 04/2026 – Present

- Facilitate high-level technical communication between faculty leads to accelerate the development of computer vision applications for clinical detection of Mild Cognitive Impairment (MCI).
- Authored and managed the Institutional Review Board (IRB) application, ensuring all research protocols met rigorous ethical and safety compliance standards for human-subject testing.
- Execute complex data processing and quantitative analysis of large-scale survey datasets regarding care robot technology adoption.
- Analyzed survey responses to identify trends in caregiver experiences and technology adoption.
- Developed data visualizations and summary statistics to communicate findings to research collaborators.

Volunteer Experience

Outreach, Stanford STEM Nights

Stanford University | 10/2025 – Present

- Orchestrated large-scale STEM engagement events, coordinating cross-functional teams to deliver technical demonstrations, ensuring rigorous safety compliance and effective public communication.
- Organizes social media.

Vice President, Outreach Director, Girls in STEM

Northview High School | 08/2022 – 05/2025

- Each month elementary school girls come and conduct science experiments.
- Organized a yearly collaboration with neighboring high schools to create a day-long camp with science experiments.
- Gives the opportunity for young girls in elementary school to experience fun science experiments, inspiring girls to pursue STEM as a career.

Awards

- 2024 + 2025 Morley Award Recipient (~23/1161)
 - Awarded to Georgia Tech's distance math students whose final grades for linear algebra and multivariable calculus were among the top 2% of the class.